

|               |                                   |
|---------------|-----------------------------------|
| Date          |                                   |
| Team ID       |                                   |
| Project Name  | Smart Metro Ticket Booking System |
| Maximum Marks | 2 Marks                           |

# SMART METRO TICKET GENERATING SYSTEM

## Proposed Solution

### 1. Solution Summary

The proposed solution is a unified, cloud-native Smart Metro Ticket Generating System that replaces manual, paper-based ticketing with a fully digital, self-service experience. Commuters purchase and validate tickets through a mobile app, web portal, or station kiosk, pay via UPI, card, NFC or a unified digital wallet, and pass through gates using contactless QR/NFC validation — with every transaction recorded in real time for operators.

The solution is designed to be rolled out incrementally: pilot stations first validate the core purchase-to-gate flow, followed by network-wide expansion once performance and reliability benchmarks are met. This phased approach limits risk, lets the team incorporate commuter feedback before scaling, and gives Metro Operations confidence in the platform before legacy counters are reduced.

Unlike a simple point solution, the proposed platform is architected to extend beyond ticketing over time — the same wallet and identity layer can later support parking, retail partnerships, or multi-modal transfers to buses and ride-share, making this investment a foundation rather than a one-off upgrade.

#### 1.1 Core Capabilities

Seven core capabilities sit at the center of the proposed solution, spanning the full commuter journey from ticket purchase through gate validation to post-trip analytics for operators.

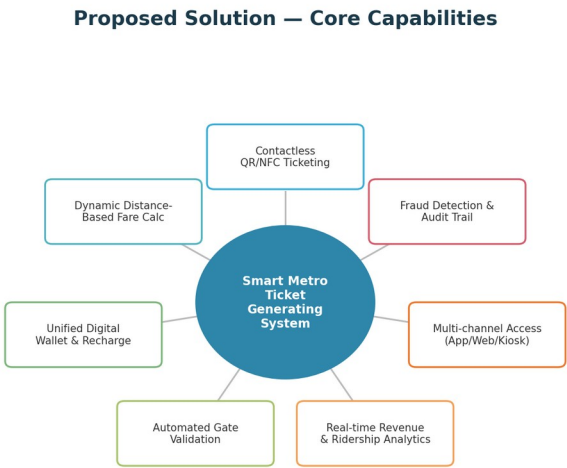


Figure 1: Proposed solution — core capabilities radiating from the central ticketing platform

## 2. What Changes: Current State vs. Proposed Solution

The table and diagram below contrast today's manual, counter-driven process against the proposed digital experience, making explicit what each stakeholder group gains from the transition. This comparison was validated against the pain points documented in the Problem–Solution Fit analysis, ensuring every proposed change traces back to an observed commuter or operator frustration rather than a generic technology refresh.

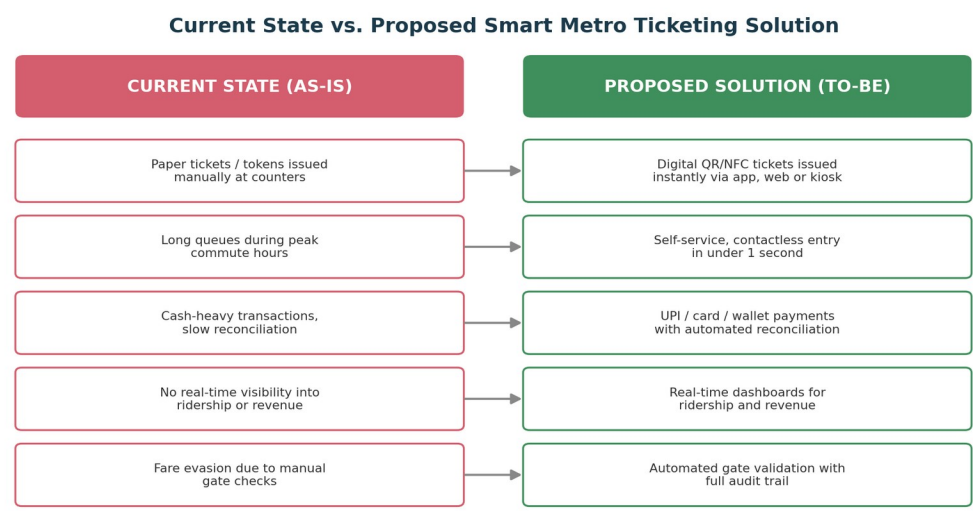


Figure 2: Current state (as-is) mapped against the proposed solution (to-be) across five key pain points

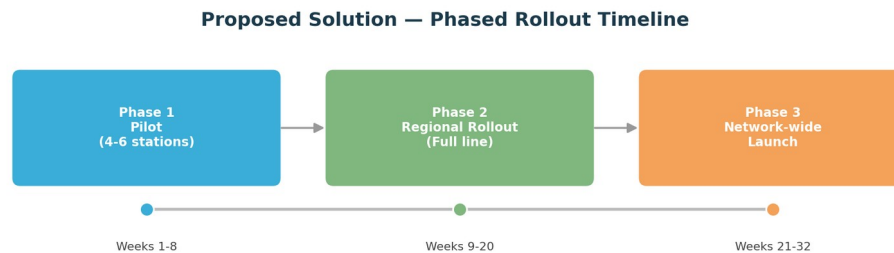
### 2.1 Feature-to-Benefit Mapping

| Feature                       | Commuter Benefit                 | Operator Benefit              |
|-------------------------------|----------------------------------|-------------------------------|
| Contactless QR/NFC ticketing  | No queues; tap-and-go entry      | Higher gate throughput        |
| Dynamic distance-based fare   | Pays only for distance travelled | Fairer, auditable pricing     |
| Unified digital wallet        | One balance across all channels  | Reduced cash handling         |
| Real-time analytics dashboard | Faster service issue resolution  | Data-driven capacity planning |
| Multi-channel access          | Buy via app, web or kiosk        | Lower counter staffing needs  |

### 3. Rollout Approach and Success Measures

The proposed solution will be delivered in three phases to manage risk and gather feedback before full network rollout:

- Phase 1 — Pilot (4-6 stations): validate ticket purchase, payment, QR/NFC issuance and gate integration end to end; gather commuter feedback.
- Phase 2 — Regional rollout: expand to a full line, integrate with existing legacy turnstile hardware, and onboard station staff to the new gate-override workflow.
- Phase 3 — Network-wide launch: roll out across all stations, decommission manual counters where volumes allow, and hand off ongoing operations to the Metro Operations and IT teams.



*Figure 3: Phased rollout timeline from pilot to network-wide launch*

#### 3.1 Success Measures

Success will be tracked against the same KPIs used to build the business case: average commuter wait time, gate throughput, fare-leakage percentage, and revenue-reconciliation time. Baseline current-state figures and proposed-solution targets for each of these are detailed in the accompanying Problem–Solution Fit document.

#### 3.2 Summary

By digitizing ticket purchase, payment and gate validation into a single connected platform, the proposed solution directly removes the friction points commuters experience today while giving Metro Operations the real-time visibility it currently lacks — setting up a foundation that can later extend to multi-modal transport integration and loyalty programs.